Making use of multivariate distributions

ADVANCED PROBABILITY: UNCERTAINTY IN DATA



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The role of probability and distributions in business decisions

- Quantify uncertainty
- Assess risks
- Better data-driven choices



Course overview

Chapter 1

Advanced probability for business decisions

Chapter 2

Measuring and managing uncertainty

Chapter 3

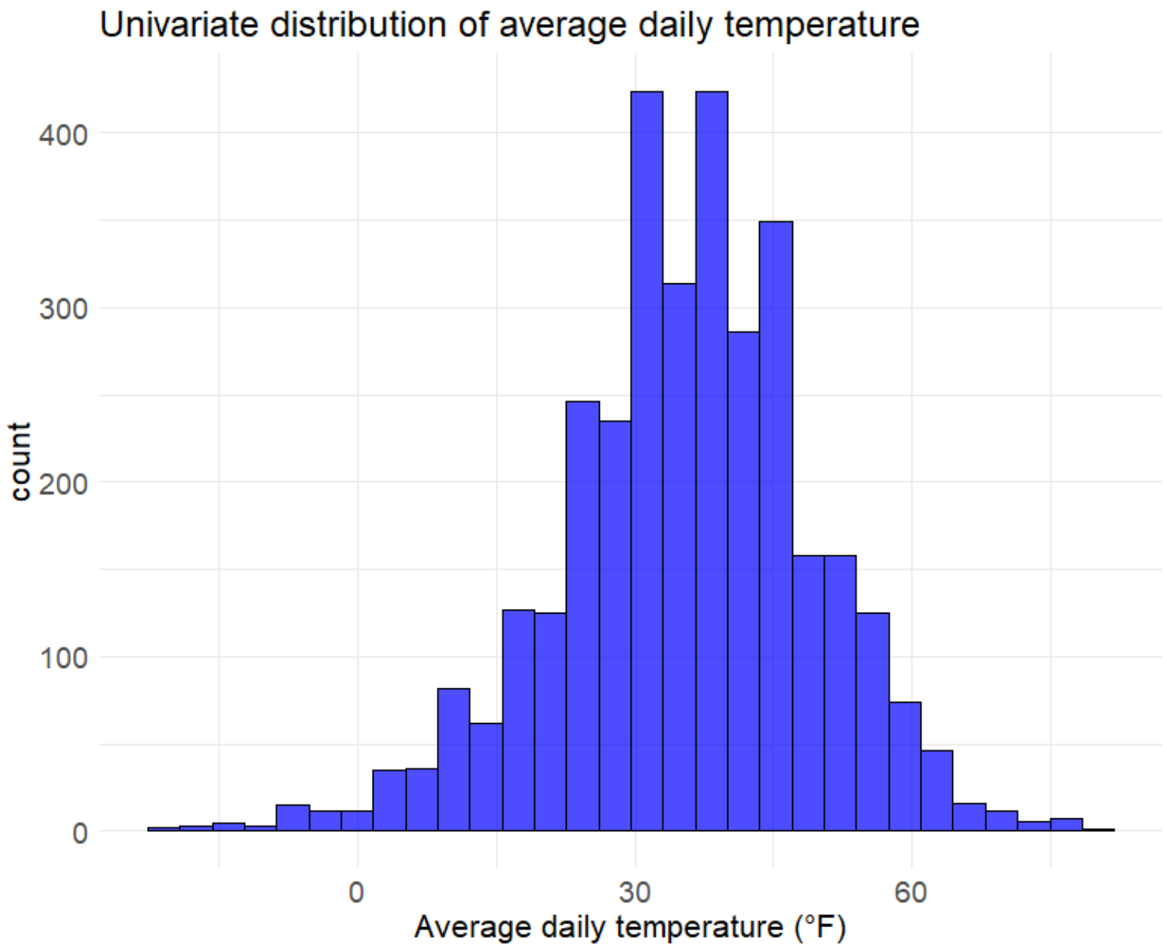
Simulation techniques for decision support

Prerequisite knowledge and skills

- Basic knowledge of statistics
- Recommended DataCamp course: **Introduction to Statistics**

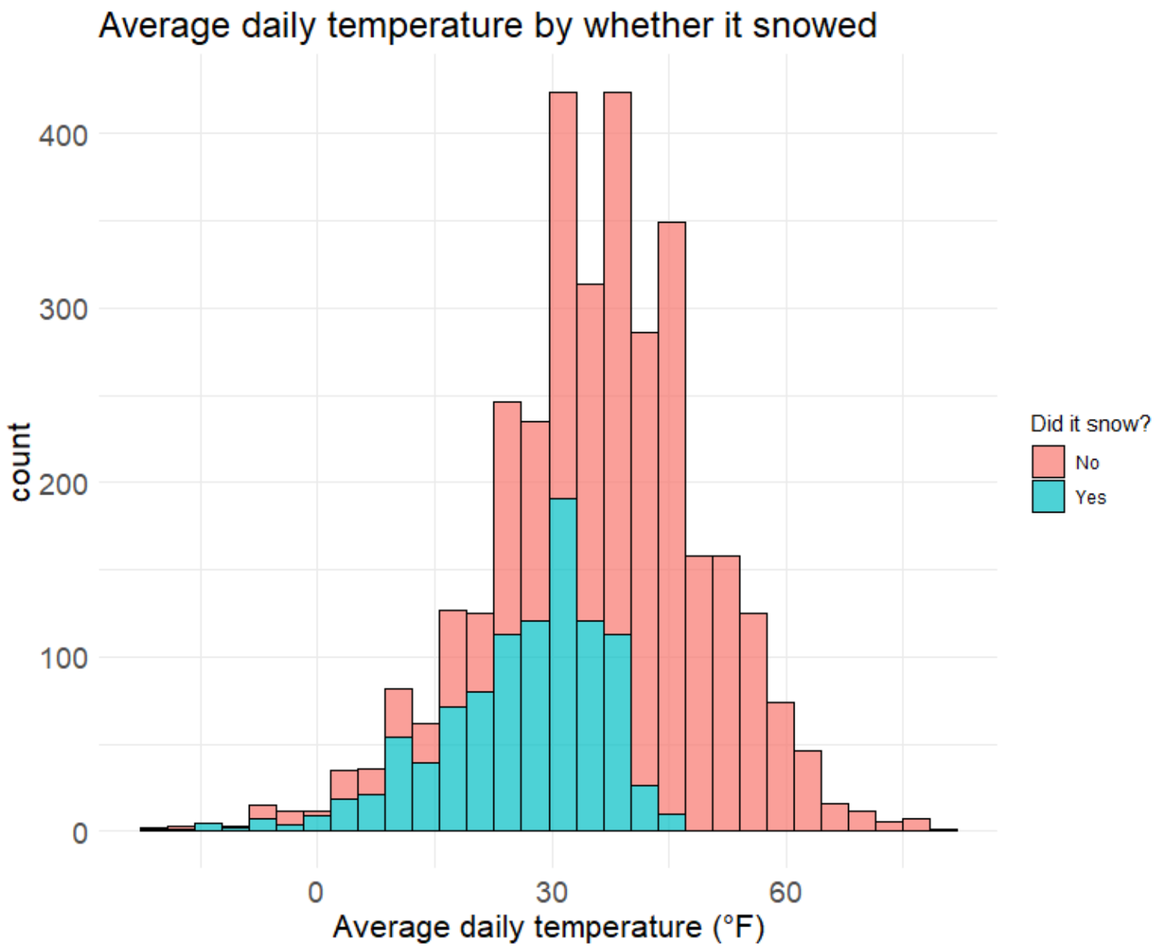
Univariate distributions

- Probabilities for a single variable
- Summarize data and make informed decisions
- However, real-world data is often more complex



Multivariate distributions

- Two or more variables interacting and changing together
- Deeper insights into data



Joint probability

- Value combinations of different variables at the same time
- How likely are they to happen simultaneously?



Example: coffee shop

- Understanding customer behavior
  - How much do customers spend?
  - How long do they stay?
- Separately:
  - Average spend: $5
  - Average stay: 30 minutes
- Together:
  - Customers who stay more than one hour spend 30% more



Example: coffee shop

Time Spent in Shop	Average Spending ($)
< 30 minutes	$5.00
30 - 60 minutes	$6.00
> 60 minutes	$6.50

- Use to plan interventions: discounts, comfortable seating

Let's practice!

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Applying conditional probability

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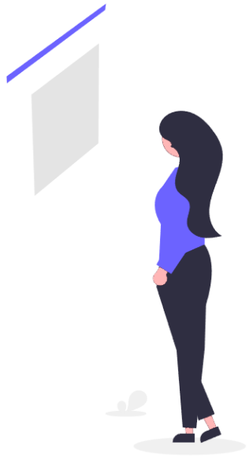


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Understanding conditional probability

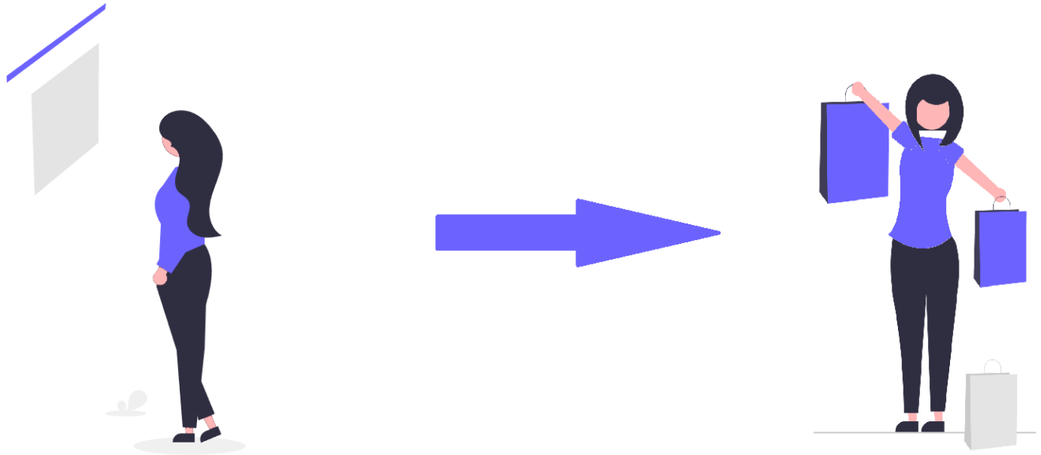
- Probability of event B occurring given that event A already happened
- Example:
  - Customer viewed product page (event A)



EVENT A

Understanding conditional probability

- Probability of event B occurring given that event A already happened
- Example:
  - Customer viewed product page (event A)
  - Will customer make purchase? (event B)



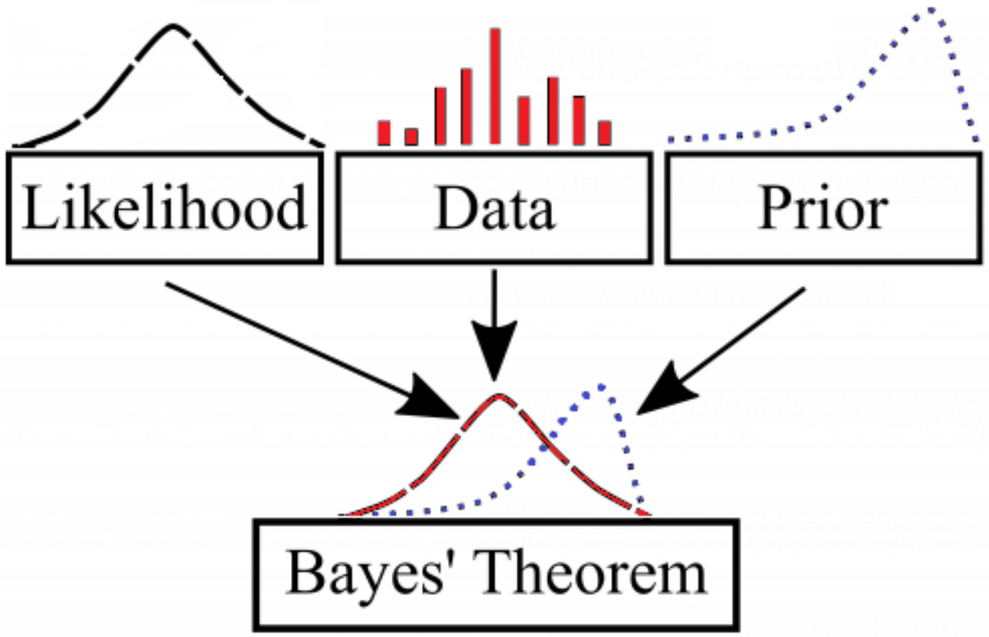
Real-world applications of conditional probability

- Customer retention analysis:
  - Will a customer renew given their use of premium features?
- Fraud detection:
  - Is a transaction fraudulent given unusual spending patterns?
- Supply chain optimization:
  - Chance of delay given bad weather?



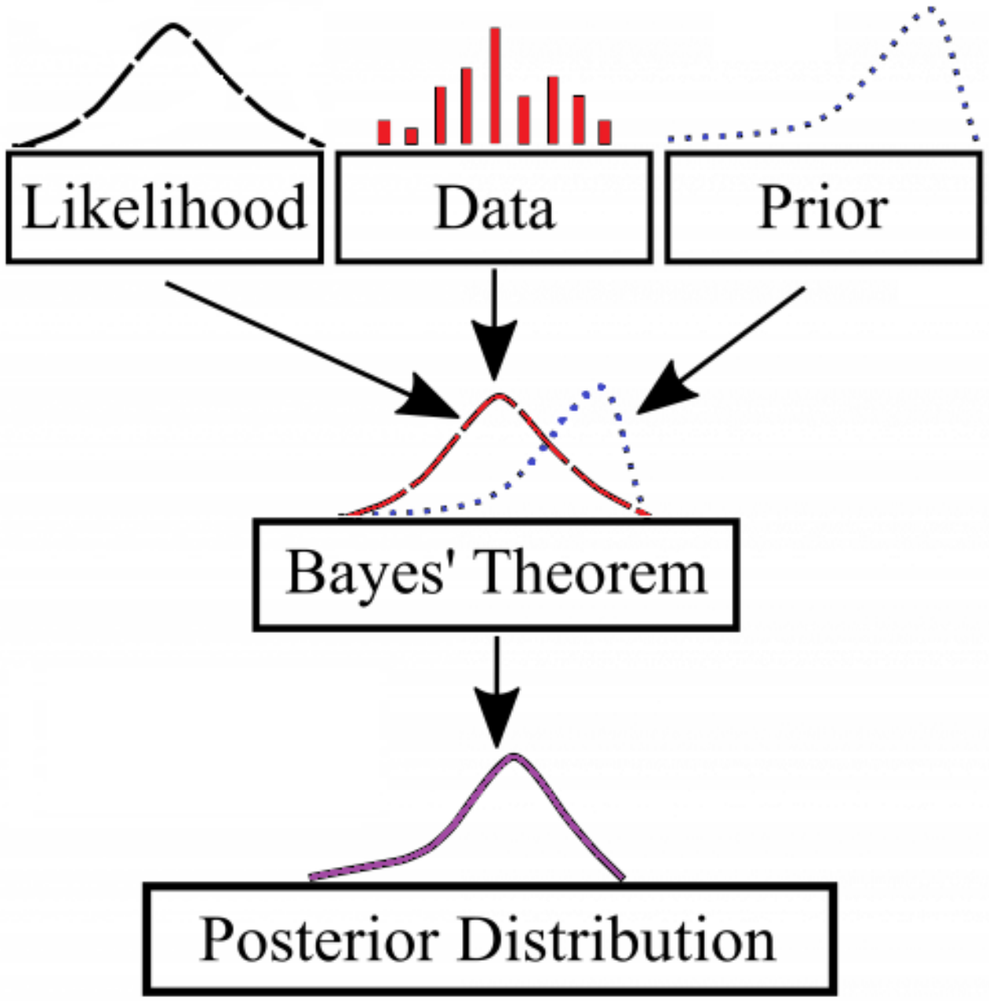
Revising probabilities with Bayes' theorem

- Revise probabilities in light of new evidence
- Start with: prior probabilities + new data + likelihood



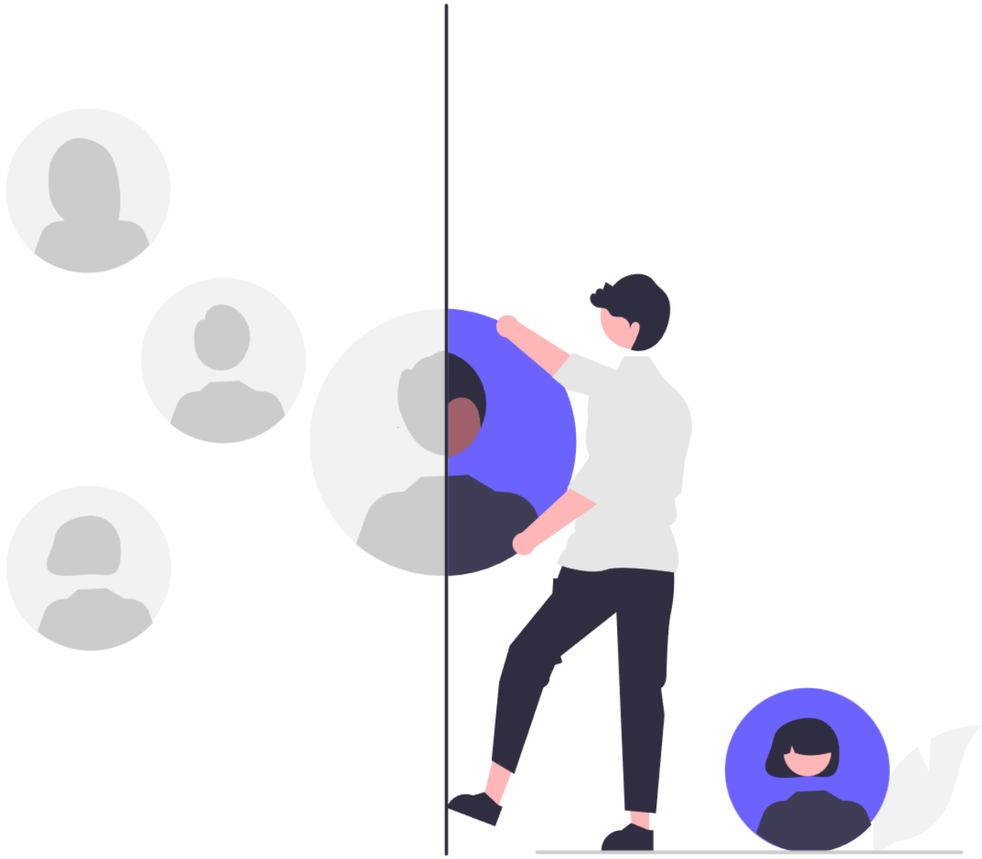
Revising probabilities with Bayes' theorem

- Revise probabilities in light of new evidence
- Start with: prior probabilities + new data + likelihood
- End with: posterior probabilities



Bayesian modeling and decision-making

- Continuous updates as new data comes available
- Manage uncertainty and incomplete data
- Adapt and improve decision-making



Example: fraud detection

- Bayesian model to track credit card fraud
  - Prior probability based on industry data: `2%`
- New transaction
  - Sudden spending change behavior
    - Present in `85%` of fraudulent transactions
    - Only in `5%` of legitimate transactions
  - Updated posterior probability rises significantly, to `25.8%`
- Transaction is flagged for review



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Markov Chain analysis

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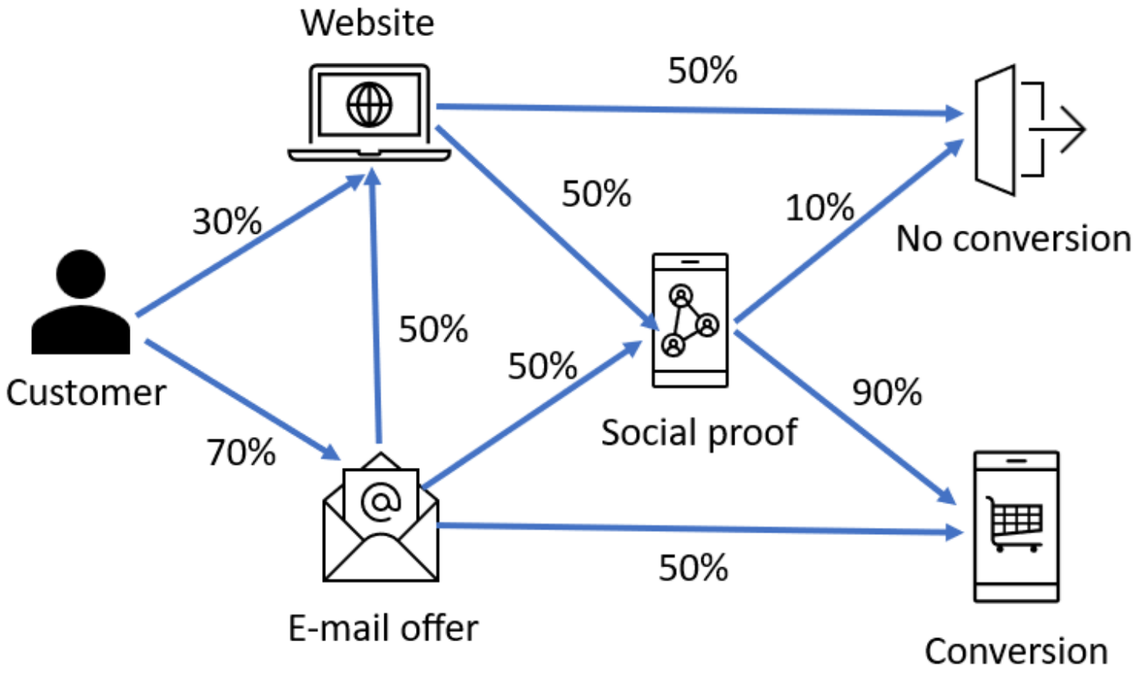


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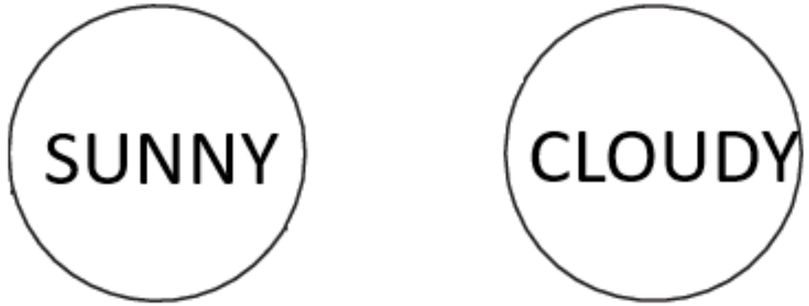
What are Markov Chains?

- Model systems with different states
- Transition between states based on probabilities
- Key idea: transition only depends on current state
- Key application: understanding customer journeys



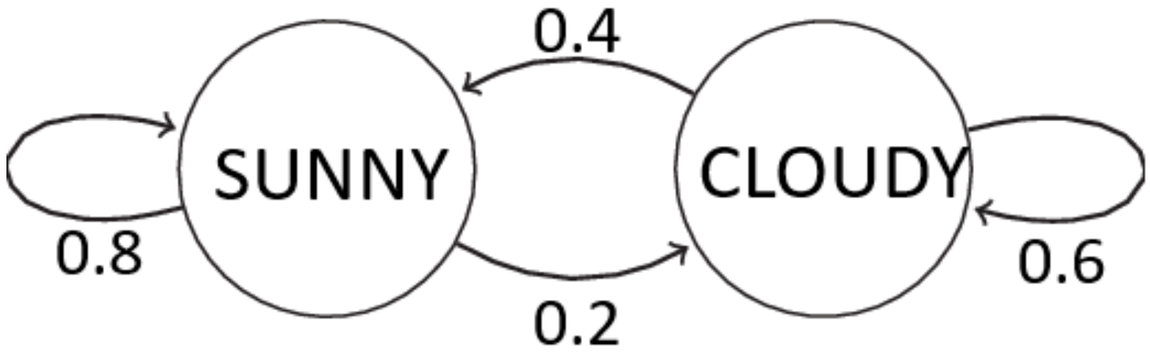
The elements of a Markov Chain

- **States:** different conditions or stages



The elements of a Markov Chain

- **States:** different conditions or stages
- **Transitions:** probabilities of moving from one state to another



The elements of a Markov Chain

- **Steady-state probabilities**: long-term probabilities of being in a state
  - Calculated using statistical software or BI tools
  - E.g. steady-state for sunny: 66.67%

Transitioning matrix

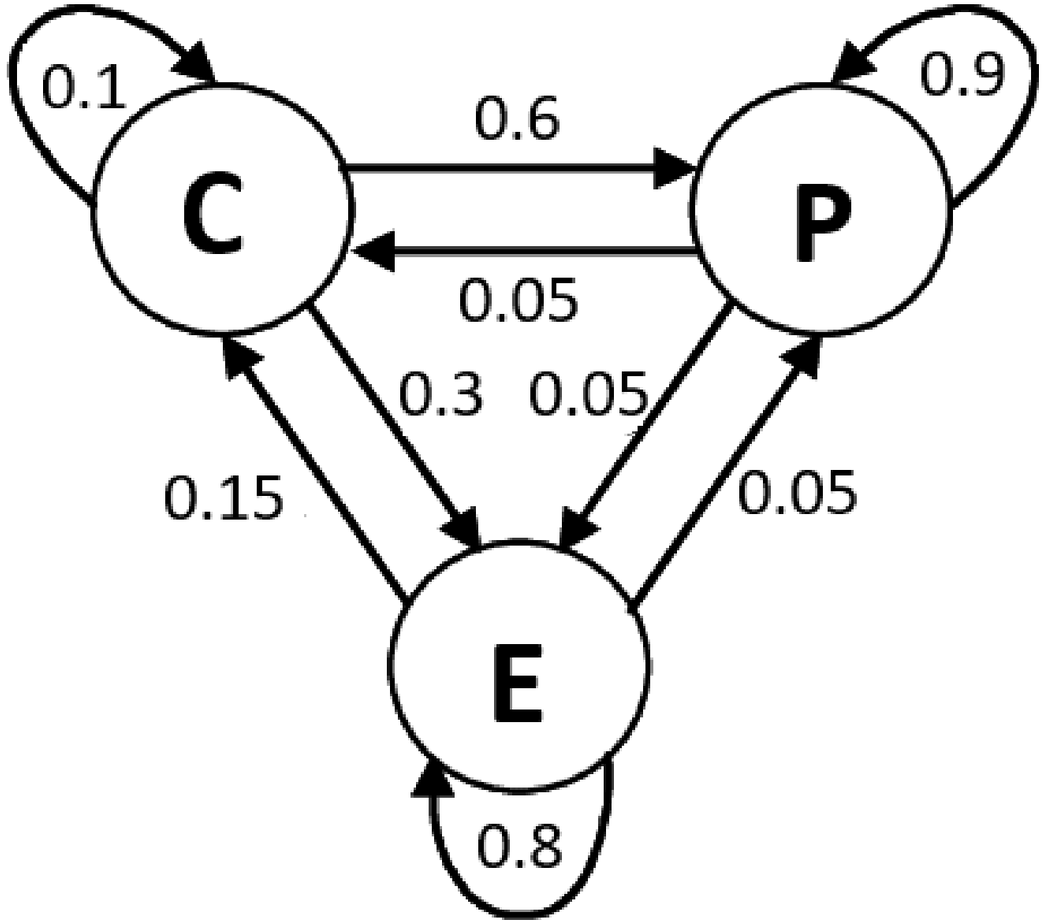
	Sunny	Cloudy
Sunny	80%	20%
Cloudy	40%	60%

Applications of Markov Chains

- Customer retention:
  - Identifying transition probabilities between engagement and churn states
- Conversion optimization:
  - Progression through a sales funnel
- Product recommendation:
  - Predicting future customer interactions



Example: e-commerce[1]



[1] C for Coupon, P for Purchase, E for Exit

Example: e-commerce

	Used a (new) coupon	Made a (new) purchase	Exited
Used a coupon	10%	60%	30%
Made a purchase	5%	90%	5%
Exited	15%	5%	80%

Example: e-commerce

	Used a (new) coupon	Made a (new) purchase	Exited
Used a coupon	10%	**60%**	**30%**
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